

ECE 3441 Spring 2022

Exam 1

Version A

Please do not open booklet unless instructed to do so, or substantial penalty will result.

- Closed book, closed notes
- ONE 8.5 x 11" double-sided cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 80 minutes.

First Name Last Name,

Peoplesoft ID,

Date *DD/MM/YYYY*

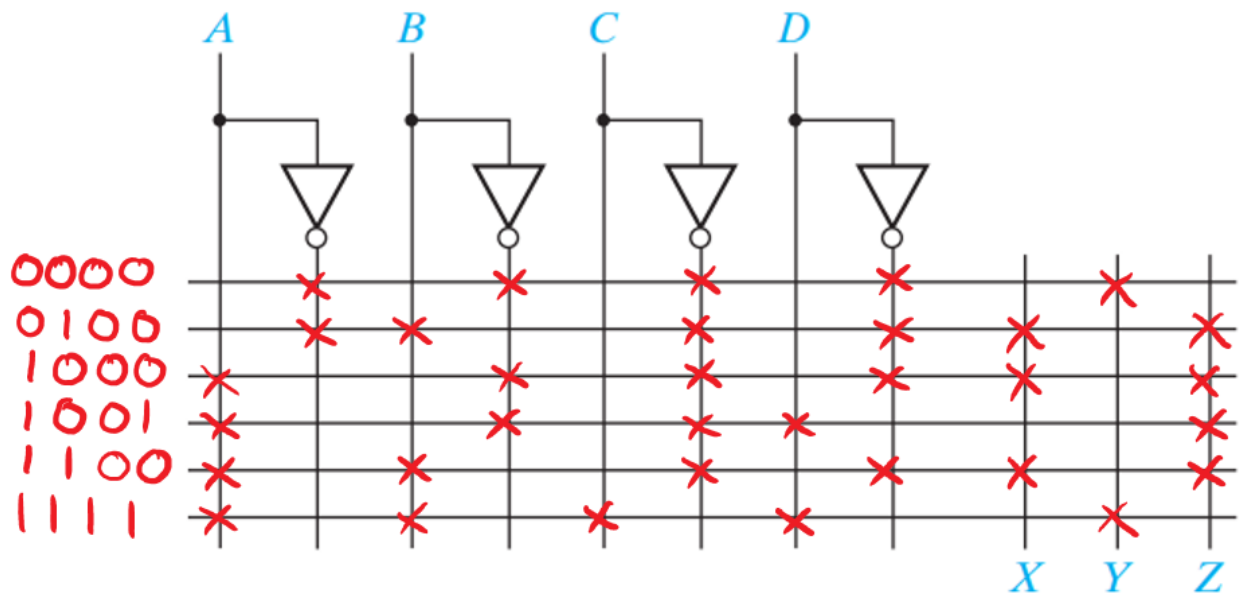
Question	Points	Maximum
1		20
2		15
3		15
4		20
5		20
6		20

Total: _____/100 possible points + 10 Bonus

Question 1:

Implement the following PLA to realize 3 number detectors (X, Y, Z). Each detector accepts a 4-bit binary input and works as follow:

- X outputs 1 if the decimal-equivalent value of the input is divisible by 4 (0 is not divisible by 4)
- Y outputs 1 if the input includes all 0s or all 1s.
- Z outputs 1 if the input contains the sequence 100, (*e.g. 100_ etc.*)



$$X = \sum m(4, 8, 12)$$

$$Y = \sum m(0, 15)$$

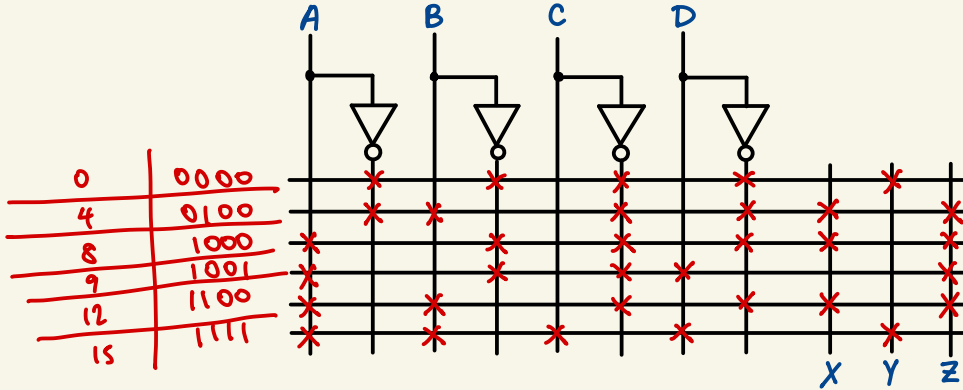
$$Z = \sum m(4, 8, 9, 12)$$

Question 1:

Implement the following PLA to realize 3 number detectors (X, Y, Z). Each detector accepts a 4-bit binary input and works as follow:

- X outputs 1 if the decimal-equivalent value of the input is divisible by 4 (0 is not divisible by 4)
- Y outputs 1 if the input includes all 0s or all 1s.
- Z outputs 1 if the input contains the sequence 100, (e.g. 100_ etc.)

AB \ CD	00		01		11		10	
	m ₀	m ₁	m ₂	m ₃	m ₄	m ₅	m ₆	m ₇
00	0000	0001	0011	0010				
01	0100	0101	0111	0110				
11	1100	1101	1111	1110				
10	1000	1001	1011	1010				



$$X = \sum m(4, 8, 12)$$

$$Y = \sum m(0, 15)$$

$$Z = \sum m(4, 8, 9, 12)$$

$$0, 4, 8, 9, 12, 15$$

Question 2:

Simplify the following expression using Boolean Algebra, identify each theorem you use.

$$G(A, B, C, D, E, F) = [(AB) \oplus (A + B)](B \oplus C) + (E \equiv F)(D \equiv E)(F \oplus D) + (A \oplus B)(B \equiv C)$$

You must use Boolean Algebra to receive any credit

$$\begin{aligned} & [(AB) \oplus (A + B)](B \oplus C) \\ &= [(AB)(A'B') + (A' + B')(A + B)](B \oplus C) \\ &= (A'B + AB')(B \oplus C) \\ &= (A \oplus B)(B \oplus C) \end{aligned}$$

$$\begin{aligned} & (E \equiv F)(D \equiv E)(F \oplus D) \\ &= (E'F + EF')(D'E + DE')(D'F + DF') \\ &= (DE'F + D'EF')(D'F + DF') \\ &= 0 \end{aligned}$$

$$\begin{aligned} G &= (A \oplus B)(B \oplus C) + 0 + (A \oplus B)(B \equiv C) \\ G &= (A \oplus B)[(B \oplus C) + (B \equiv C)] \\ G &= A \oplus B \end{aligned}$$

Simplify the following expression using Boolean Algebra, identify each theorem you use.

$$G(A, B, C, D, E, F) = [(AB) \oplus (A + B)](B \oplus C) + (E \equiv F)(D \equiv E)(F \oplus D) + (A \oplus B)(B \equiv C)$$

You must use Boolean Algebra to receive any credit

Exclusive OR

$$X \oplus Y = X'Y + XY'$$

Equivalence

$$(X \equiv Y) = XY + X'Y'$$

Equivalence $\rightarrow$
Exclusive OR



$$\begin{aligned} & [(AB)'(A+B) + (AB)(A+B)'](B'C + BC') + \cancel{(EF + E'F')(DE + D'E')(F'D + FD')} + (A'B + AB')(BC + B'C') \\ & [(A'+B')(A+B) + (AB)(A'B')] (B'C + BC') + (A'B C + \cancel{A'B B' C'} + \cancel{A B B' C} + AB'C') \quad \text{law of comp} \\ & [(\cancel{A A'} + \cancel{A' B} + \cancel{A B'} + \cancel{B B'}) + (\cancel{A A' B} + \cancel{A B' B})] (B'C + BC') \\ & (A'B + AB')(B'C + BC') + A'BC + AB'C \\ & \cancel{A' B B' C} + \cancel{A' B C'} + \cancel{A B' C} + \cancel{A B B' C} + \underline{A'BC} + AB'C \\ & B(A'C' + A'C) + B'(AC + AC) \end{aligned}$$

Question 3:

- Convert the following number from binary to base 16: **11010110.1001₂**
- Treating the binary numbers below as 2's complement. Perform the 6-bit addition and indicate overflow/no overflow.

$$\mathbf{110001 + 011011}$$

$$\begin{array}{|c|c|c|c|c|c|c|} \hline 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ \hline 13 & = & D & & 6 & & 9 & \\ \hline \end{array} \cdot \begin{array}{|c|c|c|c|} \hline 1 & 0 & 0 & 1 \\ \hline 9 & & & \\ \hline \end{array} \subscript{2} = D6.9_{16}$$

$$\begin{array}{r} 110001 \quad -15 \\ + 011011 \quad 27 \\ \hline (1)001100 \quad 12 \\ \text{no overflow} \end{array}$$

a) Convert the following number from binary to base 16: **11010110.1001₂**

b) Treating the binary numbers below as 2's complement. Perform the 6-bit addition and indicate overflow/no overflow.

$$110001 + 011011$$

$$\begin{array}{r} 543210 \\ 10+8+2+1 \\ 24+2+1 = 27 \end{array}$$

$$\begin{array}{|c|c|c|} \hline 1101 & 0110 & 1001 \\ \hline 13 & 6 & 9 \\ \hline \end{array}$$

$$13 = D$$

$$\boxed{D6.9}$$

$$\begin{array}{r} 110001 \\ + 011011 \\ \hline 1001100 \end{array} \rightarrow 12$$

$$\begin{array}{r} 110001 \\ \leftarrow 001110 \\ + \quad \quad 1 \\ \hline 001111 \end{array}$$

$$\begin{array}{r} -15 \\ +24 \\ \hline 12 \end{array}$$

$$\begin{array}{r} 543210 \\ 8421 \\ \hline 12 \quad 14 \quad 15 \end{array}$$

There is no overflow

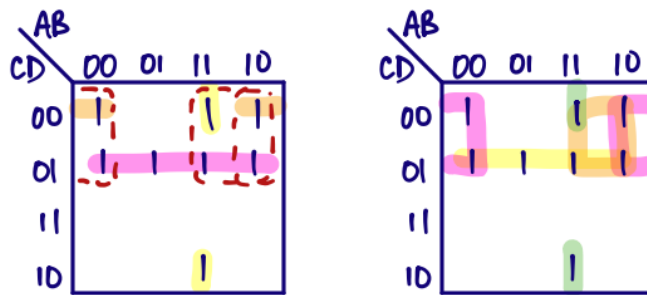
Question 4:

Given:

$$F_{1A}(A, B, C, D) = C'D + ABD' + B'C'D'$$

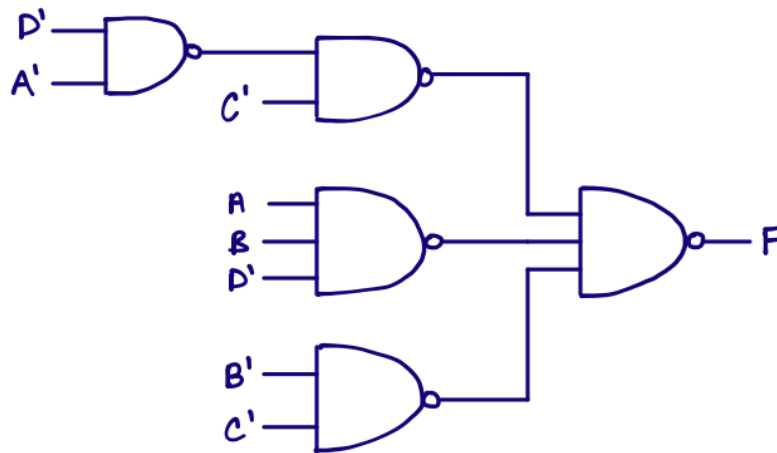
- Use a K-Map to identify all 1 hazards.
- Give the new function that has no 1 hazards.
- Realize the no hazard function using 3-input NAND gates.

$$C'D + ABD' + B'C'D'$$



$$\text{no hazards: } C'D + AC' + ABD' + B'C'$$

$$C'(D + A) + ABD' + B'C'$$



Given:

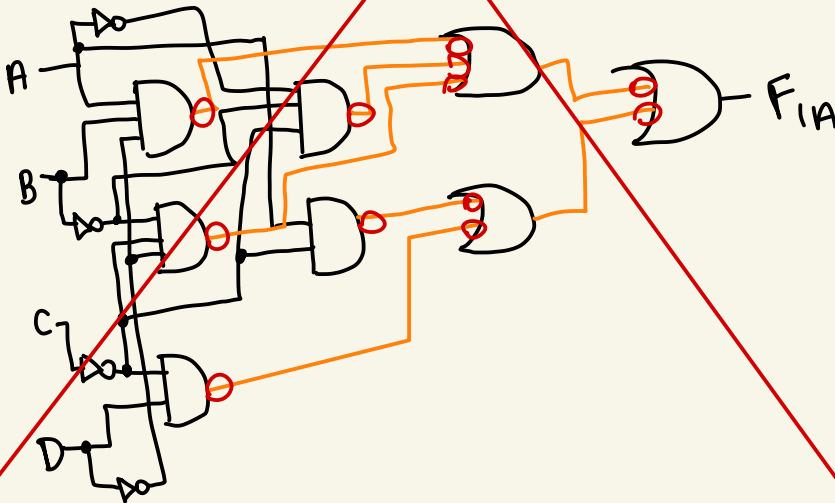
$$F_{1A}(A, B, C, D) = C'D + ABD' + B'C'D'$$

- Use a K-Map to identify all 1 hazards.
- Give the new function that has no 1 hazards.
- Realize the no hazard function using 3-input NAND gates.

AB \ CD	00	01	11	10
00	m_0 1	m_1 1	m_3 0	m_2 0
01	m_4 0	m_5 1	m_7 0	m_6 0
11	m_{12} 1	m_{13} 1	m_{15} 0	m_{14} 1
10	m_8 1	m_9 1	m_{11} 0	m_{10} 0

$$\begin{aligned} C'D & 01 \\ ABD' & 110 \\ B'C'D' & 000 \end{aligned}$$

$$F_{1A} = C'D + ABD' + B'C'D' + A'B'C' + AC'$$



Question 5:

Given:

$$F_{2A}(W, X, Y, Z) = \prod M(1, 5, 11, 12, 14, 15) \cdot \prod d(3, 4, 10)$$

- Use K-maps to find the minimum product-of-sums (POS) of F_B .
- Use K-maps to find the minimum sum-of-products (SOP) of F_B . (*You may use the same K-map as from part a*)
- Realize the function using 3-input NOR gates.

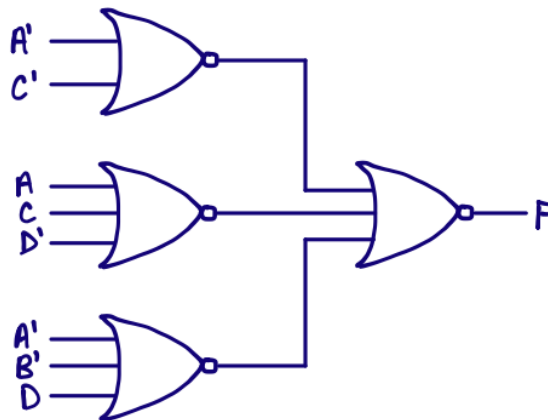
		WX			
YZ		00	01	11	10
00			X	0	
01		0	0		
11		X		0	0
10				0	X

$$F_B = (A' + C') \cdot (A + C + D') \cdot (B' + C + D)$$

or

$$F_B = (A' + C') \cdot (A + C + D') \cdot (A' + B' + D) \quad F_B = B'D' + A'C + AC'D$$

		WX			
YZ		00	01	11	10
00		1	X		1
01				1	1
11		X	1		
10		1	1		X



Given:

$$F_{2A}(W, X, Y, Z) = \prod M(1, 5, 11, 12, 14, 15) \cdot \prod d(3, 4, 10)$$

- Use K-maps to find the minimum product-of-sums (POS) of F_B .
- Use K-maps to find the minimum sum-of-products (SOP) of F_B . (You may use the same K-map as from part a)
- Realize the function using 3-input NOR gates.

POS

AB \ CD	00	01	11	10
00	m_0 0	m_1 1	m_3 X	m_2 0
01	m_4 X	m_5 1	m_7 0	m_6 0
11	m_{12} 1	m_{13} 0	m_{15} 1	m_{14} 1
10	m_8 0	m_9 0	m_{11} 1	m_{10} X

SOP

AB \ CD	00	01	11	10
00	m_0 0	m_1 1	m_3 X	m_2 0
01	m_4 X	m_5 1	m_7 0	m_6 0
11	m_{12} 1	m_{13} 0	m_{15} 1	m_{14} 1
10	m_8 0	m_9 0	m_{11} 1	m_{10} X

Question 6:

Given:

$$F_{3A}(A, B, C) = \sum m(0,4) + \sum d(1,5)$$

Derive a minimum SOP equation for F_A using Quine-McCluskey method.

Column I			Column II			Column III		
0	0	000	0	0, 1 0, 4	00- -00	0	0,1,4,5	-0-
1	1 4	001 100	1	1, 5 4, 5	-01 10-	1		
2	5	101	2					
3								

Prime implicants:

0,1,4,5

-0-

B'

$$\Rightarrow F_A = B'$$

Given:

$$F_A(A, B, C) = \sum m(0, 4) + \sum d(1, 5)$$

$\begin{matrix} & & 1 & 1 & 1 \\ & & 4 & 2 & 1 \end{matrix}$

Derive a minimum SOP equation for F_A using Quine-McCluskey method.

0	000	0,1 00-	0,4 -00
1	001	1,5 -01	1,4 10-
4	100		
5	101		

0,1,4,5 -0-

$$F_A = B'$$